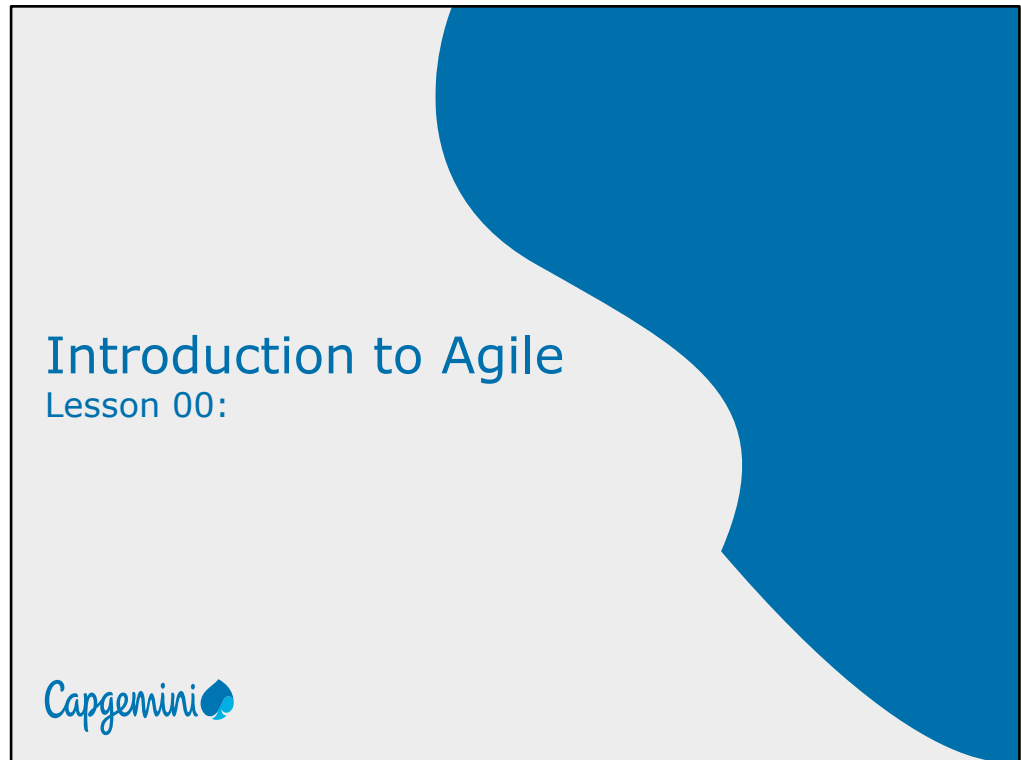




Document History



Date	Course Version No.	Software Version No.	Developer /SME	Reviewer /s	Approver	Change Record Remarks
April, 2018	1.0	NA	Shilpa Bhosle			New course created based on suggestions received from BU SMEs during Curriculum Revision Workshop

Course Goals and Non Goals



Course Goals

- Introducing participants to the Agile Software Development Model
- Introduction to Agile Practices & Methods
- Understanding SCRUM
- At the end of this program, participants gain an understanding of how to transition sound traditional test practices into an Agile Development Environment
- Understand the key differences between traditional and Agile Testing Practices
- Understand the roles and responsibilities of a typical Agile Testing Team



Course Non Goals

- This course does not cover other than anything the course goals

Pre-requisites



Basic knowledge of Software Development Life Cycle
Basic knowledge of Programming Concepts
Basic Knowledge of Software Testing Fundamentals

Intended Audience

Novice Developers
Test Engineers



Day Wise Schedule



Day 1

- Lesson 1: Agile Process Framework
- Lesson 2: Agile Methods and Practices - SCRUM

Day 2

- Lesson 3: Agile Methods and Practices - Extreme Programming (XP), Lean Software Development & Kanban
- Lesson 4: Introduction to Agile Testing
- Lesson 5: Agile Testing Quadrants and Agile Test Planning

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Lesson 1: Agile Process Framework

- 1.1. History of Traditional Software Development Model
- 1.2 Agile Software Development – Definition
- 1.3 Agile Development Model
- 1.4 Graphical Illustration of Agile Development Model
- 1.5 Why use Agile?
- 1.6 Agile Manifesto and Principles
- 1.7 12 Principles of Agile Methods
- 1.8 Agile Values

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Lesson 1: Agile Process Framework (Cont.)

- 1.15 What is NOT an Agile software development?
- 1.16 Foundation of an Agile software development Method
- 1.17 Common Characteristics of Agile Methods
- 1.18 Agile Methods and Practices
- 1.19 When to use Agile Model?
- 1.20 Advantages of Agile Model
- 1.21 Disadvantages of Agile Model
- 1.22 Difference between Agile and Waterfall Model
- 1.23 Agile – Myths and Reality
- 1.24 Agile Market Insight

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Lesson 2: Agile Methods and Practices - SCRUM

- 2.1 Introduction to SCRUM
- 2.2 Scrum Roles and Responsibilities
- 2.3 Scrum Core Practices and Artifacts
 - 2.3.1 User Story
 - 2.3.2 Sprint
 - 2.3.3 Release Planning Meeting
 - 2.3.4 Sprint Planning Meeting
 - 2.3.5 Daily Scrum Meeting (Daily Stand up)
 - 2.3.6 Sprint Review Meeting
 - 2.3.7 Retrospective
 - 2.3.8 Product Backlog
 - 2.3.9 Sprint Backlog
 - 2.3.10 Burn-Down Chart
 - 2.3.11 Velocity

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Lesson 2: Agile Methods and Practices – SCRUM (Cont.)

- 2.3.12 Impediment Backlog
- 2.16 Definition of “Done”
- 2.17 Splitting User Story into Task
- 2.18 Why to Split User Story into Task?
- 2.19 Guidelines for Breaking Down a User Story into Tasks
- 2.20 Examples of Scrum Task Board
- 2.21 Planning Poker®
- 2.22 Planning Poker® - Process/Steps
- 2.23 What are Story Points?
- 2.24 How do We Estimate in Story Points?
- 2.25 What Goes into Story Points?

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Lesson 3: Agile Methods and Practices - Extreme Programming (XP), Lean Software Development & Kanban

- 3.1 Introduction to Extreme Programming
- 3.2 The Rules of Extreme Programming
- 3.3 Extreme Programming (XP) - Principles
- 3.4 Extreme Programming (XP) – Key Terms
- 3.5 Introduction to Lean Software Development
- 3.6 Principles of Lean Software Development
- 3.7 What is Kanban?

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Lesson 4: Introduction to Agile Testing

- 4.1 What is Agile Testing?
- 4.2 Agile Team - Roles and Activities
- 4.3 Where does Tester fit in Agile Team?
- 4.4 Agile Team – Tester’s Role and Responsibilities
- 4.5 Agile Team – Test Manager’s Role and Responsibilities
- 4.6 How is Agile Testing different?
- 4.7 Traditional Testing Vs. Agile Testing
- 4.8 What is Iteration 0?
- 4.9 User Story Perspective Agile Testing Process
- 4.10 Tester’s Change in Mind-Set – A key to success

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Lesson 5: Agile Testing Quadrants and Agile Test Planning

- 5.1 An overview of Agile Testing Quadrants
- 5.2 Agile Testing Quadrant 1, 2 & 3 Goals
- 5.3 Agile Testing Quadrant 1, 2 & 3 Toolkit
- 5.4 Test Planning in Agile Testing

References



Websites :

- www.extremeprogramming.org
- www.agilemanifesto.org
- www.wikipedia.org
- www.scrum-institute.org
- www.agilealliance.org
- www.agilemodeling.com
- www.scrumguides.org
- www.mountangoatsoftware.com

Books :

- Agile Testing: A Practical Guide for Testers and Agile Teams - Lisa Crispin, Janet Gregory
- User Stories Applied: For Agile Software Development – Mike Cohn

Other Parallel Technology Areas



None

Next Step Courses



None